NOTES:

AN ELECTRONICALLY SIGNED AND REGISTERED INSTALLATION CERTIFICATE (CF2R) POSTED BY THE INSTALLING CONTRACTOR SHALL BE SUBMITTED TO THE FIELD INSPECTOR DURING CONSTRUCTION AT THE BUILDING SITE. A REGISTERED CF2R WILL HAVE A UNIQUE 21-DIGIT REGISTRATION NUMBER FOLLOWED BY FOUR ZEROS LOCATED AT THE BOTTOM OF EACH PAGE. THE FIRST 23 DIGITS OF THE NUMBER WILL MATCH THE REGISTRATION NUMBER OF THE ASSOCIATED CF2R. CERTIFICATE OF OCCUPANCY WILL NOT BE ISSUED UNTIL FORMS CF2R IS REVIEWED AND APPROVED

AN ELECTRONICALLY SIGNED AND REGISTERED CERTIFICATE(S) OF FIELD VERIFICATION AND DIAGNOSTIC TESTING (CF3R) SHALL BE POSTED AT THE BUILDING SIGNED AND REGISTERED CERTIFICATE(S) OF FIELD VERIFICATION AND DIAGNOSTIC TESTING (CF3R) SHALL BE POSTED AT THE BUILDING SITE BY A CERTIFIED HERS RATER. A REGISTERED CF3R WILL HAVE A UNIQUE 25-DIGIT REGISTRATION NUMBER LOCAED AT THE BOTTOM OF EACH PAGE. THE FIRST 20 DIGITS OF THE NUMBER WILL MATCH THE REGISTRATION NUMBER OF THE ASSOCIATED CF2R. CERTIFICATE OF OCCUPANCY WILL NOT BE ISSUED UNTIL CF3R IS REVIEWED AND APPROVED

MANDATORY MEASURES SUMMARY: Residential (Page 1 of 3) MF-1R

Project Name	Date

NOTE: Low-rise residential buildings subject to the Standards must comply with all applicable mandatory measures listed, regardless of the compliance approach used. More stringent energy measures listed on the Certificate of Compliance (CF-1R, CF-1R-ADD, or CF-1R-ALT Form) shall supersede the items marked with an asterisk (*) below. This Mandatory Measures Summary shall be incorporated into the permit documents, and the applicable features shall be considered by all parties as minimum component performance specifications whether they are shown elsewhere in the documents or in this summary. Submit all applicable sections of the MF-1R Form with plans.

Building Envelope Measures:

§110.6(a)1: Doors and windows between conditioned and unconditioned spaces are manufactured to limit air leakage.

§110.6(a)4: Fenestration products (except field-fabricated windows) have a label listing the certified U-Factor, certified Solar Heat Gain Coefficient (SHGC), and infiltration that meets the requirements of §10-111(a).

§110.7: Exterior doors and windows are weather-stripped; all joints and penetrations are caulked and sealed.

§110.8(a): Insulation specified or installed meets Standards for Insulating Material. Indicate type and include on CF-2R Form.

§110.8(i): The thermal emittance and solar reflectance values of the cool roofing material meets the requirements of §110.8(i) when the installation of a Cool Roof is specified on the CF-1R Form.

*§150.0(a): Minimum R-30 (R-19 for Additions/Alterations) insulation in wood-frame ceiling or equivalent U-factor.

§150.0(b): Loose fill insulation shall conform with manufacturer's installed design labeled R-Value.

*§150.0(c): Minimum R-13 insulation in 2x4 wood-frame wall (R-19 in 2x6) or equivalent U-factor.

*§150.0(d): Minimum R-19 insulation in raised wood-frame floor or equivalent U-factor.

§150.0(f): Air retarding wrap is tested, labeled, and installed according to ASTM E1677-95(2000) when specified on the CF-1R Form.

§150.0(g): Mandatory Vapor barrier installed in Climate Zones 14 or 16.

§150.0(l): Water absorption rate for slab edge insulation material alone without facings is no greater than 0.3%; water vapor permeance rate is no greater than 2.0 perm/inch and shall be protected from physical damage and UV light deterioration.

§150.0(q) Fenestration Products. Fenestration separating conditioned space from unconditioned space or outdoors shall meet the requirements of either Item 1 or 2 below:
1. Fenestration, including skylight products, must have a maximum U-factor of 0.58.
2. The weighted average U-factor of all fenestration, including skylight products, shall not exceed 0.58.

EXCEPTION to Section 150.0(q)1: Up to 10 square feet of fenestration area or 0.5 percent of the Conditioned Floor Area, whichever is greater, is exempt from the maximum U-factor requirement.

§150.0(r) Solar Ready Buildings. Shall meet the requirements of Section 110.10 applicable to the building project.

Fireplaces, Decorative Gas Appliances and Gas Log Measures:

§150.0(e)1A: Masonry or factory-built fireplaces have a closable metal or glass door covering the entire opening of the firebox.

§150.0(e)1B: Masonry or factory-built fireplaces have a combustion outside air intake, which is at least six square inches in area and is equipped with a with a readily accessible, operable, and tight-fitting damper and or a combustion-air control device.

§150.0(e)2: Continuous burning pilot lights and the use of indoor air for cooling a firebox jacket, when that indoor air is vented to the outside of the building, are prohibited.

Space Conditioning, Water Heating and Plumbing System Measures:

§110.0-§110.3: HVAC equipment, water heaters, showerheads, faucets and all other regulated appliances are certified by the Energy Commission.

§110.3(c)5: Water heating recirculation loops serving multiple dwelling units and High-Rise residential occupancies meet the air release valve, backflow prevention, pump isolation valve, and recirculation loop connection requirements of §110.3(c)5.

§110.5: Continuously burning pilot lights are prohibited for natural gas: fan-type central furnaces, household cooking appliances (appliances with an electrical supply voltage connection with pilot lights that consume less than 150 Btu/hr are exempt), and pool and spa heaters.

§150.0(h): Heating and/or cooling loads are calculated in accordance with ASHRAE, SMACNA or ACCA.

§150.0(i): Heating systems are equipped with thermostats that meet the setback requirements of Section 110.2(c).

§150.0(j)1A: Storage gas water heaters rated with an Energy Factor no greater than the federal minimal standard are externally wrapped with insulation having an installed thermal resistance of R-12 or greater.

§150.0(j)1B: Unfired storage tanks, such as storage tanks or backup tanks for solar water-heating system, or other indirect hot water tanks have R-12 external insulation or R-16 internal insulation where the internal insulation R-value is indicated on the exterior of the tank.

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§150.0(j)2A: All domestic hot water system piping conditions listed below, whether buried or unburied, must be insulated per TABLE 120.3-A.
i. The first 5 feet (1.5 meters) of hot and cold water pipes from the storage tank.
ii. All piping with a nominal diameter of 3/4 inch (19 millimeter) or larger.
iii. All piping associated with a domestic hot water recirculation system regardless of the pipe diameter.
iv. Piping from the heating source to storage tank or between tanks.
v. Piping buried below grade.
vi. All hot water pipes from the heating source to the kitchen fixtures.

§150.0(j)2: Pipe insulation for steam hydronic heating systems or hot water systems >15 psi, meets the requirements of Standards Table 120.3-A.

§150.0(j)3A: Insulation is protected from damage, including that due to sunlight, moisture, equipment maintenance, and wind.

§150.0(j)4: Solar water-heating systems and/or collectors are certified by the Solar Rating and Certification Corporation.

§150.0(m)1: All air-distribution system ducts and plenums installed, are sealed and insulated to meet the requirements of CMC Sections 601, 602, 603, 604, 605 and Standard 6-5; supply-air and return-air ducts and plenums are insulated to a minimum installed level of R-6 or enclosed entirely in conditioned space. Openings shall be sealed with mastic, tape or other duct-closure system that meets the applicable requirements of UL 181, UL 181A, or UL 181B or aerosol sealant that meets the requirements of UL 723. If mastic or tape is used to seal openings greater than 1/4 inch, the combination of mastic and either mesh or tape shall be used

§150.0(m)1: Building cavities, support platforms for air handlers, and plenums defined or constructed with materials other than sealed sheet metal, duct board or flexible duct shall not be used for conveying conditioned air. Building cavities and support platforms may contain ducts. Ducts installed in cavities and support platforms shall not be compressed to cause reductions in the cross-sectional area of the ducts.

§150.0(m)2D: Joints and seams of duct systems and their components shall not be sealed with cloth back rubber adhesive duct tapes unless such tape is used in combination with mastic and draw bands.

§150.0(m)7: Exhaust fan systems have back draft or automatic dampers.

§150.0(m)8: Gravity ventilating systems serving conditioned space have either automatic or readily accessible, manually operated dampers.

§150.0(m)9: Insulation shall be protected from damage, including that due to sunlight, moisture, equipment maintenance, and wind. Cellular foam insulation shall be protected as above or painted with a coating that is water retardant and provides shielding from solar radiation that can cause degradation of the material.

§150.0(m)10: Flexible ducts cannot have porous inner cores.

§150.0(n)1: Systems using gas or propane water heaters, whether tank or on-demand, to serve individual dwelling units shall include all the following components :
A. A 120V electrical receptacle that is within 3 feet from the water heater and accessible to the water heater with no obstructions;
B. A Category III or IV vent, or a Type B vent with straight pipe between the outside termination and the space where the water heater is installed;
C. A condensate drain that is no more than 2 inches higher than the base of the installed water heater, and allows natural draining without pump assist,
D. A gas supply line with a capacity of at least 200,000 Btu/hr.

§150.0(o): All dwelling units shall meet the requirements of ANSI/ASHRAE Standard 62.2 Ventilation and Acceptable Indoor Air Quality in Low-Rise Residential Buildings. Window operation is not a permissible method of providing the Whole Building Ventilation required in Section 4 of that Standard.

Pool and Spa Heating Systems and Equipment Measures:

§110.4(a): Any pool or spa heating system shall be certified to have: a thermal efficiency that complies with the Appliance Efficiency Regulations; an on-off switch mounted outside of the heater; a permanent weatherproof plate or card with operating instructions; and shall not use electric resistance heating or a pilot light.

§110.4(b)1: Any pool or spa heating equipment shall be installed with at least 36" of pipe between filter and heater, or dedicated suction and return lines, or built-up connections for future solar heating.

§110.4(b)2: Outdoor pools or spas that have a heat pump or gas heater shall have a cover.

§110.4(b)3: Pools shall have directional inlets that adequately mix the pool water, and a time switch that will allow all pumps to be set or programmed to run only during off-peak electric demand periods.

§150.0(p): Residential pool systems or equipment meet the pump sizing, flow rate, piping, filters, and valve requirements of §150.0(p).

Residential Lighting Measures:

§150.0(k)1A: Installed luminaires shall be classified as high-efficacy or low-efficacy for compliance with Section 150.0(k) in accordance with TABLE 150.0-A or TABLE 150.0-B, as applicable.

§150.0(k)1C: The wattage of permanently installed luminaires shall be determined as specified by §130.0(c).

§150.0(k)1D: Ballasts for fluorescent lamps rated 13 Watts or greater shall be electronic and shall have an output frequency <= 20 kHz.

§150.0(k)1E: Permanently installed night lights and night lights integral to installed luminaires or exhaust fans shall be rated to consume no more than five watts of power per luminaire or exhaust fan as determined in accordance with Section 130.0(c). Night lights shall not be required to be controlled by vacancy sensors.

MANDATORY MEASURES SUMMARY: Residential (Page 3 of 3) MF-1R

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§150.0(k)1F: Lighting integral to exhaust fans, in rooms other than kitchens, shall meet the applicable requirements of §150.0(k).

§150.0(k)2: All switching devices and controls shall meet the requirements of §150.0(k)2.

§150.0(k)3: A minimum of 50 percent of the total rated wattage of permanently installed lighting in kitchens shall be high efficacy.
EXCEPTION: Up to 50 watts for dwelling units less than or equal to 2,500 ft² or 100 watts for dwelling units larger than 2,500 ft² may be exempt from the 50 percent high efficacy requirement when all lighting in the kitchen is controlled in accordance with the applicable provisions in Section 150.0(k)2, and is also controlled by vacancy sensors or dimmers.

§150.0(k)4: Permanently installed lighting that is internal to cabinets shall use no more than 20 watts of power per linear foot of illuminated cabinet.

§150.0(k)5: Lighting installed in bathrooms shall meet the following requirements:
A. A minimum of one high efficacy luminaire shall be installed in each bathroom; and
B. All other lighting installed in each bathroom shall be high efficacy or controlled by vacancy sensors.

§150.0(k)6: Lighting installed in attached and detached garages, laundry rooms, and utility rooms shall be high efficacy luminaires and controlled by vacancy sensors.

§150.0(k)7: Lighting installed in rooms or areas other than in kitchens, bathrooms, garages, laundry rooms, and utility rooms shall be high efficacy, or shall be controlled by either dimmers or vacancy sensors.
EXCEPTION 1: Luminaires in closets less than 70 square feet.
EXCEPTION 2: Lighting in detached storage building less than 1000 square feet located on a residential site.

§150.0(k)8: Luminaires recessed into insulated ceilings shall be listed for zero clearance insulation contact (IC) by Underwriters Laboratories or other nationally recognized testing/rating laboratory; and have a label that certifies the luminaire is airtight with air leakage less than 2.0 CFM at 75 Pascals when tested in accordance with ASTM E283; and be sealed with a gasket or caulk between the luminaire housing and ceiling.

§150.0(k)9A: For single-family residential buildings, outdoor lighting permanently mounted to a residential building or other buildings on the same lot shall be high efficacy, or may be low efficacy if it meets all of the following requirements:
- i. Controlled by a manual ON and OFF switch that does not override to ON the automatic actions of items ii or iii below; and
- ii. Controlled by a motion sensor not having an override or bypass switch that disables the motion sensor, or controlled by a motion sensor having a temporary override switch which temporarily bypasses the motion sensing function and automatically reactivates the motion sensor within 6 hours
- iii. Controlled by one of the following methods:
  - a. Photocontrol not having an override or bypass switch that disables the photocontrol; or
  - b. Astronomical time clock not having an override or bypass switch that disables the astronomical time clock, and which is programmed to automatically turn the outdoor lighting OFF during daylight hours; or
  - c. Energy management control system which meets all of the following requirements:
    At a minimum provides the functionality of an astronomical time clock in accordance with Section 110.9; meets the Installation Certification requirements in Section 130.4; meets the requirements for an EMCS in Section 130.5; does not have an override or bypass switch that allows the luminaire to be always ON; and, is programmed to automatically turn the outdoor lighting OFF during daylight hours.

§150.0(k)9A: For low-rise multi-family residential buildings, outdoor lighting for private patios, entrances, balconies, and porches; and outdoor lighting for residential parking lots and residential carports with less than eight vehicles per site shall comply with one of the following requirements:
i. Shall comply with Section 150.0(k)9A; or
ii. Shall comply with the applicable requirements in Sections 110.9, 130.0, 130.2, 130.4, 140.7, and 141.0.
§150.0(k)9: For low-rise residential buildings with four or more dwelling units, outdoor lighting not regulated by Section 150.0(k)9B or Section 150.0(k)9D shall comply with the applicable requirements in Sections 110.9, 130.0, 130.2, 130.4, 140.7, and 141.0.
§150.0(k)9D: Outdoor lighting for residential parking lots and residential carports with a total of eight or more vehicles per site shall comply with the applicable requirements in Sections 110.9, 130.0, 130.2, 130.4, 140.7, and 141.0.

§150.0(k)10: Internally illuminated address signs shall comply with Section 140.8; OR not contain a screw-base socket, and consume no more than five watts of power as determined according to §130.0(d).

§150.0(k)11: Lighting for residential parking garages for eight or more vehicles shall comply with the applicable requirements for nonresidential garages in Sections 110.9, 130.0, 130.1, 130.4, 140.6, and 141.0.

§150.0(k)12A. In a low-rise multi-family residential building where the total interior common area in a single building equals 20 percent or less of the floor area, permanently installed lighting for the interior common areas in that building shall be high efficacy luminaires or controlled by an occupant sensor.
§150.0(k)12B. In a low-rise multi-family residential building where the total interior common area in a single building equals more than 20 percent of the floor area, permanently installed lighting in that building shall:
i. Comply with the applicable requirements in Sections 110.9, 130.0, 130.1, 140.6, and 141.0; and
ii. Lighting installed in corridors and stairwells shall be controlled by occupant sensors that reduce the lighting power in each space by at least 50 percent. The occupant sensors shall be capable of turning the light fully On and Off from all designed paths of ingress and egress.

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